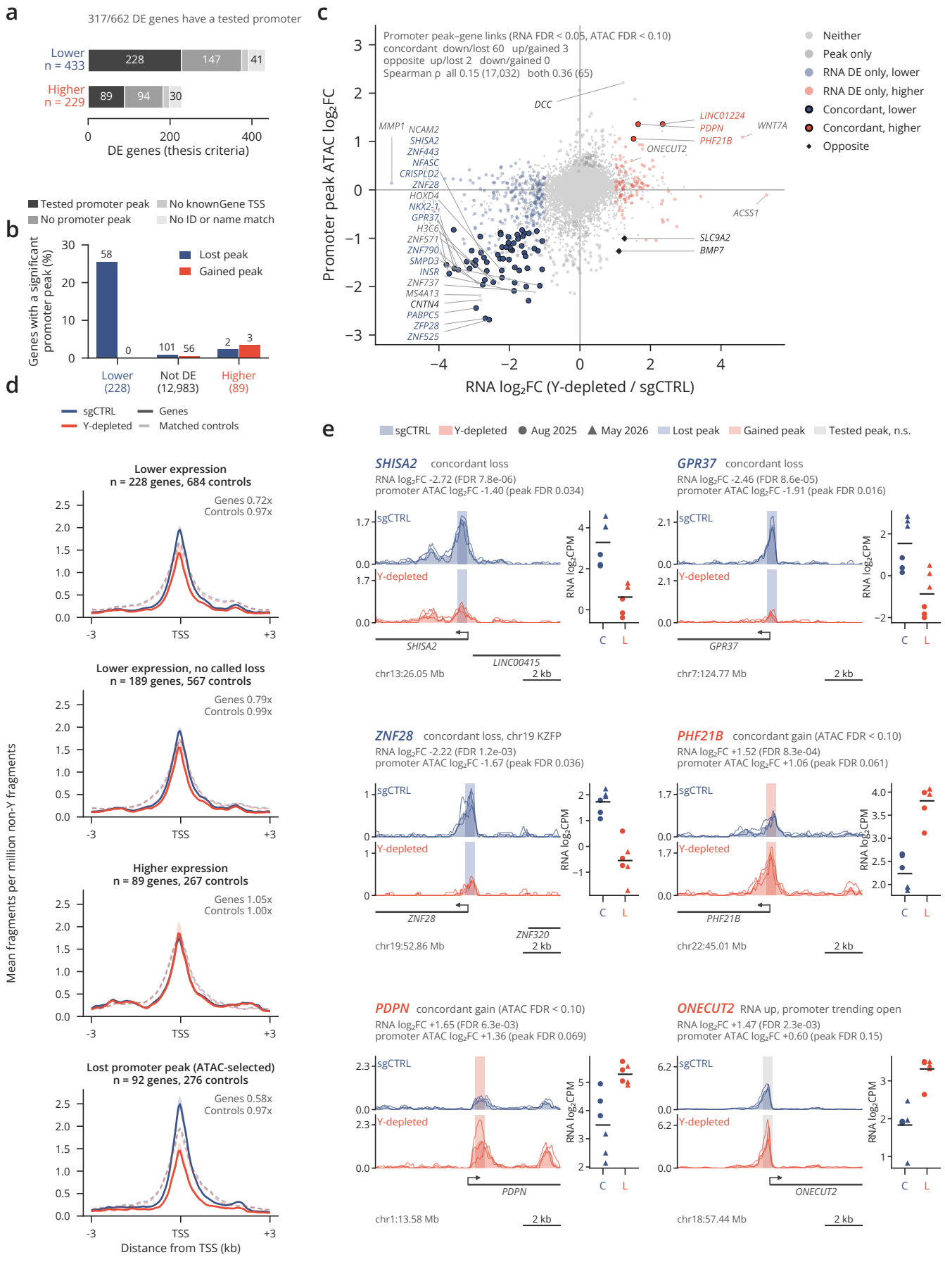

a
317/662 DE genes have a tested promoter
Lower n = 433
228
147
41
Higher n = 229
89
94
30
0
200
400
DE genes (thesis criteria)
b
Tested promoter peak
No knownGene TSS
No promoter peak
No ID or name match
Genes with a significant promoter peak (%)
Lost peak
Gained peak
58
0
101
56
2
3
Lower (228)
Not DE (12,983)
Higher (89)
c
Promoter peak–gene links (RNA FDR < 0.05, ATAC FDR < 0.10)
concordant down/lost 60 up/gained 3
opposite up/lost 2 down/gained 0
Spearman ρ all 0.15 (17,032) both 0.36 (65)
Promoter peak ATAC log2FC
RNA log2FC (Y-depleted / sgCTRL)
DCC
MMP1
NCAM2
SHISA2
ZNF443
NFASC
CRISPLD2
ZNF28
HOXD4
NKX2-1
GPR37
H3C6
ZNF571
ZNF790
SMPD3
INSR
ZNF737
MS4A13
CNTN4
PABPC5
ZFP28
ZNF525
LINC01224
PDPN
PHF21B
WNT7A
ONECUT2
ACSS1
SLC9A2
BMP7
Neither
Peak only
RNA DE only, lower
RNA DE only, higher
Concordant, lower
Concordant, higher
Opposite
d
sgCTRL
Y-depleted
Genes
Matched controls
Lower expression
n = 228 genes, 684 controls
Genes 0.72x
Controls 0.97x
Lower expression, no called loss
n = 189 genes, 567 controls
Genes 0.79x
Controls 0.99x
Higher expression
n = 89 genes, 267 controls
Genes 1.05x
Controls 1.00x
Lost promoter peak (ATAC-selected)
n = 92 genes, 276 controls
Genes 0.58x
Controls 0.97x
Mean fragments per million non-Y fragments
-3
TSS
+3
Distance from TSS (kb)
e
sgCTRL
Y-depleted
Aug 2025
May 2026
Lost peak
Gained peak
Tested peak, n.s.
SHISA2 concordant loss
RNA log2FC -2.72 (FDR 7.8e-06)
promoter ATAC log2FC -1.40 (peak FDR 0.034)
GPR37 concordant loss
RNA log2FC -2.46 (FDR 8.6e-05)
promoter ATAC log2FC -1.91 (peak FDR 0.016)
ZNF28 concordant loss, chr19 KZFP
RNA log2FC -2.22 (FDR 1.2e-03)
promoter ATAC log2FC -1.67 (peak FDR 0.036)
PHF21B concordant gain (ATAC FDR < 0.10)
RNA log2FC +1.52 (FDR 8.3e-04)
promoter ATAC log2FC +1.06 (peak FDR 0.061)
PDPN concordant gain (ATAC FDR < 0.10)
RNA log2FC +1.65 (FDR 6.3e-03)
promoter ATAC log2FC +1.36 (peak FDR 0.069)
ONECUT2 RNA up, promoter trending open
RNA log2FC +1.47 (FDR 2.3e-03)
promoter ATAC log2FC +0.60 (peak FDR 0.15)
RNA log2CPM
C
L
chr13:26.05 Mb
chr7:124.77 Mb
chr19:52.86 Mb
chr22:45.01 Mb
chr1:13.58 Mb
chr18:57.44 Mb
2 kb